8 **(a)** Make p the subject of

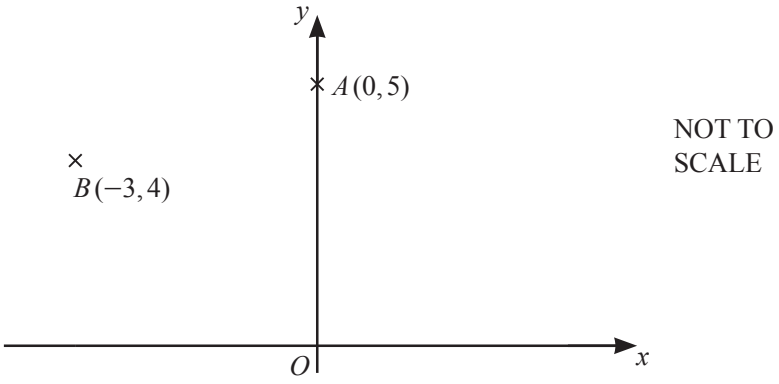
(i) $5p + 7 = m$,

$p =$ ..... [2]

(ii) $y^2 - 2p^2 = h$.

$p =$ ..... [3]

(b)



(i) Write $\overrightarrow{OA}$ as a column vector.

$\overrightarrow{OA} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

(ii) Write $\overrightarrow{AB}$ as a column vector.

$\overrightarrow{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

(iii) A and B lie on a circle, centre O.

Calculate the length of the arc AB.

..... [6]